Aparna Dharmalingam

Functional Safety | ISO 26262 | ADAS & System Safety

M.Sc. Functional Safety Engineering candidate with applied research and automotive experience in HARA, ISO 26262, SOTIF, System Safety and ADAS Safety.

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EDUCATION

M.Sc. Functional Safety Engineering
University of Kassel

10/2020 – Present — *Kassel, Germany*

- Specialization: Automotive Functional Safety
- System Safety and system-level safety concepts

Bachelor of Engineering – Mechatronics
SCSVMV University

08/2015 – 04/2019 — *Kanchipuram, India*

- Focus areas: MEMS, power electronics, control systems, and robotics

PROFESSIONAL EXPERIENCE

Student Research Assistant – Safety Engineering
Fraunhofer IESE

01/2026 – 04/2026 — *Kaiserslautern*

- Transformed free-text hazardous events from HARA artifacts into standardized YAML safety models, creating a consistent, machine-readable basis for further safety analysis.
- Hazardous events to ODD parameters and operating conditions, improving the traceability of ASIL classification.
- Developed a Python-based tool chain to automate the processing of TOD elements and hazardous-event mappings, reducing manual data-preparation effort.

Working Student – Functional Safety (ADAS & AD)
Mingabyte GmbH

03/2024 – 09/2024 — *Munich*

- Developed technical training material on automated-driving functions and ADAS Safety, enabling the structured communication of complex safety topics.
- Structured vehicle-level safety requirements and safety concepts, improving their clarity, consistency, and technical readability.
- Contributed as a co-assessor to a Functional Safety Assessment of a hydrogen fuel-cell system, supporting the systematic evaluation of safety-related evidence.

Working Student – Functional Safety
DEKRA Digital Products & Services

03/2023 – 09/2023 — *Stuttgart*

- Tested an internal Functional Safety Assessment tool against ISO 26262 and IEC 61508, identifying gaps between tool logic and applicable requirements.
- Developed standards-based checklists and review workflows, supporting more consistent and repeatable internal safety assessments.
- Documented test findings and improvement proposals, contributing to further tool development and improved usability.

CORE COMPETENCIES

Safety Standards & Methods
ISO 26262 · IEC 61508 · IEC 61131-3 · IEC 61511 · ISO/SAE 21434 · ISO 21448 (SOTIF) · HARA · FMEA · FMEDA · HAZOP · FTA · Requirements Engineering

MBSE
System-level documentation, simulation, and design

Software & Tools
Ansys medini analyze · Python · NI LabVIEW · NI Multisim · PSIM · MATLAB/Simulink · ROS/SLAM · MS Office · PTC Creo · VBA · Git · YAML · B&R Automation Studio

Personal Strengths
Analytical thinking · structured approach · initiative · teamwork

ACADEMIC THESES

Master's Thesis - In Preparation

LLM-supported HARA generation using structured safety data and evaluation against an industrial HARA.

Bachelor's Thesis – Sensor Assisted Braking System (09/2018 – 04/2019)

Modelled, developed, and validated a semi-electronic braking system on the Smart Trike hybrid vehicle; published under DOI: 10.1166/asem.2020.2606

PROJECTS

SafeStop AI - Human-in-the-Loop FMEA Pipeline
Personal Functional Safety Portfolio Project | 2026

- Built an AI-assisted preliminary FMEA workflow using controlled YAML inputs, safety requirements, GitHub Actions and mandatory human review.
- TRIAL-01: 29 AI candidates; 15/18 reference mechanisms surfaced (83.3% coverage), 12 valid additions, 1 duplicate and 93.1% usefulness precision.
- Portfolio: aparna3498.github.io/safestop-fmea-pipeline/

ADDITIONAL EXPERIENCE

Autonomous Systems Engineer
Herkules Racing Team

02/2023 – 12/2024 — *Kassel*

- Developed and tested ROS- and Python-based software modules for perception and planning on an autonomous competition vehicle.
- Supported sensor integration and safety-related troubleshooting at vehicle level.

LANGUAGES

English	German	Tamil
C1	*B1*	*C2*